

Certified AI System = Trustworthy? Exploring Expert and Lay User Perceptions and Needs Regarding AI Certification

Sarah Abdelwahab Gaballah
Ruhr University Bochum
Bochum, Germany
sarah.gaballah@ruhr-uni-bochum.de

Magdalena Wischnewski
University of Duisburg-Essen
Essen, Germany
magdalena.wischnewski@tu-dortmund.de

Nur Efsan Cetinkaya
University of Duisburg-Essen
Essen, Germany
nur.cetinkaya@uni-due.de

Martina Angela Sasse
Computer Science
Ruhr University Bochum
Bochum, Germany
martina.sasse@ruhr-uni-bochum.de

Abstract

AI certification has emerged as a promising mechanism to enhance transparency, accountability, and public trust. However, end-user perspectives remain largely unexplored. This study investigates two groups with differing AI expertise. Through qualitative interviews with 30 participants (15 experts, 15 lay users), we examined how AI certification influences trust, who should conduct it, transparency needs, post-certification monitoring, and certification fraud. Results reveal key differences between the two groups. Lay users perceive AI certification more positively than experts. Both groups prefer independent certifiers, with experts being more open to certification by private companies. Experts favor post-certification monitoring tied to system updates, whereas lay users prefer annual checks. Both groups value transparency, but the specific details they require differ. Regarding fraudulent AI certification, experts emphasize technical safeguards, while lay users focus on legal enforcement. The study discusses the implications of its findings and offers several recommendations for improving AI certification schemes.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI); HCI design and evaluation methods; User studies.**

Keywords

Artificial Intelligence, AI Governance, AI Certification, Certification Labels, Trust in AI, Calibrated Trust, Transparency

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1 Introduction

The rapid and far-reaching adoption of artificial intelligence (AI) systems in domains ranging from political and economic to social [8, 50, 60] highlights the increasing need for regulatory frameworks that can foster trust in these systems and ensure compliance with human rights and ethical standards [20]. Recognizing this need, several initiatives have introduced regulatory frameworks to establish standards for trustworthy AI, ensuring the safety, fairness, and accountability of these systems. For example, the European Union's AI Act and Brazil's AI Bill 2338/2023 both set out clear risk-based rules for AI developers to foster trustworthy AI. Across these different approaches, a common theme is the call for AI certifications [2].

Certification schemes are envisioned as tools to assess and communicate compliance with ethical and technical standards, especially in high-stakes or high-risk contexts. Despite the importance of these schemes for AI, there is limited empirical understanding of how AI certifications function in practice. First empirical studies on the effect of certifications on users' trust in AI systems show mixed results [24, 56, 62, 72]. Likewise, while certifications have long played a vital role in signaling safety and quality (see, e.g., [15, 37, 64]), AI presents fundamentally different challenges. Unlike traditional systems, AI models often operate as "black boxes," learning complex patterns from data in ways that are not always interpretable or predictable. Their performance can degrade silently over time, be highly sensitive to context, or encode hidden biases from training data [46]. Moreover, AI systems can evolve post-deployment through continuous learning or dynamic interactions with users, making static certification less helpful. Considering these challenges, the effectiveness of such certifications in fostering user trust and appropriate reliance remains an open question.

To that end, HCI research plays a critical role in both bridging a possible disconnect between regulatory ambitions and user experience and providing empirically tested information on how regulations can be effective [67]. In the present work, we aim to inform the design of certifications as user-centered AI governance tools that are not only technically robust but also socially and experientially meaningful. We pose and answer the following research questions:



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- RQ1: What are the perceptions and expectations of non-experts and AI experts regarding AI certification and post-certification monitoring, and how does transparency about these processes influence trust in AI?
- RQ2: How do these perceptions and expectations differ in the context of high-risk AI systems?
- RQ3: How do non-experts and AI experts perceive the risk of falsifying AI certifications, and what are their expectations for preventing such falsification?

The study addresses these research questions by investigating perceptions of AI certification among two user groups: those with high AI expertise (AI experts) and those with no or limited AI expertise (lay users). Considering both experts and non-experts provides a more comprehensive view of expectations: experts offer technical insights, while non-experts reflect broader public impact. Both perspectives are essential to ensure certifications are technically sound, trustworthy, and aligned with public understanding.

To answer our research questions, we conducted semi-structured interviews with 30 participants (15 AI expert users, 15 non-expert users), revealing notable differences in perspectives between the two groups. For RQ1, non-experts generally viewed certification as a trust cue, whereas experts were skeptical unless confident in the certification process. Non-experts favored transparency, seeing it as a way to increase trust, while experts worried that technical details might overwhelm lay users. Both groups preferred non-profit certifiers and highlighted the importance of post-certification monitoring, with experts favoring checks aligned with system updates and non-experts favoring periodic reviews. For RQ2, both groups agreed that high-risk AI requires stricter standards: experts emphasized mandatory, context-specific certification and real-world testing, while non-experts stressed frequent verification and stronger government oversight. For RQ3, participants expressed concern about certification fraud, such as copying legitimate labels or buying certificates without evaluation; experts prioritized technical countermeasures, whereas non-experts focused on legal enforcement. Regarding when they would verify the authenticity of a certificate an AI system has, experts stated that they would always check it before using the system for the first time, whereas non-experts mentioned that they would do so mainly when using systems they considered high-risk or suspicious.

Research contributions: This paper makes the following contributions:

- (1) We present an empirical study of how users—both AI experts and non-experts—perceive AI certification, with a particular focus on post-certification monitoring, fraud risks, and anti-fraud mechanisms. By comparing the two groups, we highlight differences in their perceptions, expectations, and concerns.
- (2) We translate these insights into actionable design and governance recommendations for AI certification that better align with users' needs, support appropriate trust calibration, and improve the communication of certification to different user groups.

2 Related Work

This section provides an overview of the relevant existing literature on AI governance and certification, with a particular emphasis on contributions from the HCI community.

2.1 Regulating AI: An HCI Perspective

With the increased integration of AI-enabled systems into everyday life, power is shifting from human decision-makers to algorithms. Because AI failures can have severe, sometimes even fatal, consequences, calls for responsible and trustworthy deployment are increasing. As an answer to the proliferation of AI, regulatory efforts have shifted from soft law to hard law. Prominent examples are the European Union's AI Act, the United States' Executive Order 14110, Canada's AI and Data Act, China's Interim Measures for Generative AI Services, and Brazil's AI Bill 2338/2023. The result is a complex, divergent, and fragmented global AI landscape [2]. While the global AI regulation landscape in itself poses various questions and problems, single initiatives themselves generate numerous difficulties. The EU AI Act (AIA), for example, has been criticized as being too ambiguous [19, 58, 66, 68], posing "interpretative challenges" [67, p. 249], or, by confusing trustworthiness with acceptability, as "overselling its regulatory ambition" [40, p. 3]. Moreover, these acts have been criticized for lending too much power to corporate players [1, 41].

Historically, HCI as an academic discipline has played a crucial role in engaging with and shaping the regulation of technologies [33]. On the one hand, the HCI community contributes to regulatory efforts by questioning the implementation of existing regulations from a human-centric perspective. Most prominently, for example, the EU AIA requires explainability as a key condition for regulatory compliance, particularly for high-risk systems. However, existing HCI research indicates that explainability may produce undesired outcomes. For instance, Eiband et al. [26] showed that placebic explanations, explanations that do not convey meaningful information, can create levels of trust similar to those generated by real explanations. In addition, Ferguson et al. [28] found that explanations can increase users' cognitive load, while Bertrand et al. [7] demonstrated that cognitive biases can impair the accurate processing of explanations. Similar examples can be found for the implementation of human oversight [39, 40, 58].

Aiming to create pathways for the HCI community to advance the implementation of regulations such as the EU AIA, Urquhart et al. [65], for example, suggest various "legal provocations" (p. 1). As such, Urquhart et al. [65] argue that the requirements of the EU AIA invite contributions from HCI scholars by, for example, designing and developing appropriate solutions to data quality requirements, risk management, technical documentation, and human oversight. Likewise, Valdez et al. [67] argue that the HCI community plays a crucial role in shaping AI regulation by bridging the gap between technical capabilities and human-centered requirements.

On the other hand, the HCI community addresses current development and design solutions, oftentimes in direct conversation with regulatory mandates or in extension to those. For example, Andalibi and Ingber [3] surveyed a U.S. representative sample, assessing participants' opinions about the deployment of systems that can infer emotions and other affective phenomena of individuals

(i.e., Emotion Artificial Intelligence). Their results show a significant discomfort across various high-stakes contexts, leading the authors to advocate against the use of Emotion AI and call for regulations. Likewise, Wang et al. [70] analyzed and reviewed age-appropriate design for children, resulting in a code for Age-Appropriate AI Design, including recommendations for privacy, manipulation and exploitation, safety and safeguarding, as well as transparency and accountability.

To conclude, the HCI community is ideally positioned to show the need for but also extend, design, and question AI regulation. Extending on this, in the next section, we review current HCI literature on how certification schemes can function as tools to assess and communicate compliance with ethical and technical standards.

2.2 Signaling Compliance with Regulation: Introducing AI Certifications

Human-computer interactions are complex interactions as they require users to infer system states, anticipate responses, and integrate multiple streams of information simultaneously. To facilitate this process, the HCI literature has come up and investigated a wide range of communicative cues that aim to help users understand what a system can do, how it should be operated, and whether it complies with specific standards.

To that end, colors as well as typography have been found to convey risks associated with system use [30]. Likewise, charts (e.g., [51]), metaphors (e.g., [18]), and icons (e.g., privacy nutrition labels [36], AI legibility marks [43]) have been found to support lay user comprehension of complex data environments and AI systems by driving attention and signaling importance. Similarly, empirical studies have examined more complex visualizations such as the Impact Assessment Card [10] or AI Failure Cards [63], finding that these visualizations communicate the risks and benefits of specific AI systems and help users find mitigation strategies.

Likewise, labels have been found to help end users gain insights into systems. For example, Caven et al. [16] show that the effectiveness of the U.S. Cyber Trust Mark was driven by demand. In their work, the authors found that, when purchasing a smart device, users were most concerned with the product pricing and least concerned with cybersecurity. Hence, Caven et al. [16] only found an effect of the Trust Mark for users who were generally security aware. Trying to increase multi-level transparency recommendations, approaches such as the AI Nutrition Facts [34], Open Ethic's Nutrition Label [27], and energy labels [29], as well as data or model cards [23, 49, 52] have been introduced and tested.

The success of interventions such as those reported above suggests that AI certifications should aid users when interacting with AI systems. Theoretical accounts of certifications and empirical insights from domains outside of AI confirm this. However, the first empirical results on the effectiveness of AI certifications are mixed.

From a theoretical perspective, certifications generally function "as a crucial mechanism for providing reliable documentation that attests to the fulfillment of predetermined standards" [13, p. 4224], aiming to reduce complexity and uncertainties about products and systems. For the context of AI, different standards have been introduced (see e.g., [4, 22, 31, 71, 74]). Once a certification has been

issued by an auditing instance (either an organization or an individual), the certification is communicated through a certification mark or seal.

To that extent, certifications are not new. Certifications and their accompanying certification marks have been introduced in various contexts such as e-commerce [37], cybersecurity [44], cloud computing [61], and health care [55]. The effectiveness of certifications has been empirically tested and mostly confirmed. In the context of online retail, for example, certifications have been shown to increase consumers' trust, outperforming alternatives such as rating systems that are linked to ambiguity, negativity, and risk [15]. Likewise, Web Assurance Seal Services (WASS) have been shown to be effective instruments to increase users' trust and mitigate user concerns about e-commerce platforms [37]. Moreover, certification labels were found to reduce perceptions of risk and to increase brand reputation [64]. However, as Riegelsberger et al. [54] have pointed out, certifications only benefit users if certifiers are both capable of performing the checks and motivated to do so, and if the seals representing passed checks are not easy to fake or subvert. In other words, certifications as indicators for trustworthiness should be costly [42].

First empirical studies in the context of AI have shown mixed results for the effectiveness of certifications. Using an established trust mark, Scharowski et al. [56] found that certifications indicated through labels significantly increased users' trust and willingness to use AI, especially for high-risk contexts. Likewise, de Almeida et al. [24] found that AI certifications increased users' trust and acceptance of AI. Additionally, the authors found that transparent communication about the certification criteria was important. Contrary to these results, Wischniewski et al. [72] found no effect of an AI trust label. The authors explain their null results that the initial level of users' trust was already high and could not be increased through the presence of an AI certification, indicating a ceiling effect. Addressing the question of who should issue a certification, Blösser and Weihrach [9] found that, over different decision domains, users prefer governmental institutions and NGOs as certifiers. Building upon this foundation, Szymczyk et al. [62] investigated the relationship between certification and explainability as two methods to enhance user trust. The 2x2 experiment revealed that explanations substantially increased both subjective and factual understanding of the AI's decisions, whereas the certification seal alone had no significant effect on trust. The absence of information regarding the issuing body or evaluation criteria of the seal is a notable point, and it is possible that this limitation contributes to the seal's limited impact.

Going beyond the direct effect on users, scholars have also tried to implement existing certification frameworks to AI systems. For example, Autischer et al. [5] documented the practical application and limitations of existing certification catalogues in the light of the EU AIA. The authors identified some limitations of the certification catalogues and proposed ideas on how to streamline the certification process. Likewise, Brogle et al. [13] conducted a case study, piloting a trial of Responsible Artificial Intelligence (RAI) certification. Their results made clear that, while the pilot provided a comprehensive evaluation of the AI system, "the audit criteria were too complex for digestibility" (p. 4235).

While these first insights exist, there are two specific aspects of certifications that have hardly been addressed in the current literature: 1) post-certification monitoring, and 2) certification fraud and anti-fraud measures. AI systems can evolve after deployment through continuous learning or interactions with users, which limits the effectiveness of static, one-time certifications. At the same time, users' trust in AI certifications may be influenced by their perceptions of the likelihood of certification fraud.

2.3 The Present Work

Despite the growing body of research on AI certifications, there remains a significant gap in understanding the specific expectations and desires of end users regarding these certifications. While prior work has focused on the effectiveness of certifications in building trust and mitigating risks, less attention has been paid to the actual perceptions and expectations of those who would be most affected by AI systems—the users themselves.

Including end users in the development of an AI certification is essential to ensure that the resulting standards are not only technically sound but also socially relevant and trustworthy. End users contribute contextual knowledge that helps surface practical concerns such as usability, accessibility, and fairness—issues often overlooked in developer- or regulator-led processes [69]. Their involvement improves the ethical and social alignment of AI systems by fostering trust [57]. Moreover, participatory approaches strengthen legitimacy and accountability by grounding abstract principles like transparency and safety in the lived experiences and expectations of those affected [3, 45, 59].

To address this gap, our study focuses on gathering insights directly from end users—including both non-experts and AI experts—to understand their expectations of AI certification. By engaging with users, we aim to inform the design, communication, and monitoring of AI certification in ways that meet the needs of different user groups and support trust calibration, helping users rely on certified systems in an informed rather than overconfident way.

3 Methodology

We conducted an interview study with 30 participants, divided into two groups: 15 AI experts and 15 non-experts (lay users), to address our research questions. Prior to conducting the study, we preregistered it on the Open Science Framework (OSF)¹. The study consisted of two parts: a drawing exercise in which participants were asked to explain and sketch their perceptions and expectations of the AI system certification process, and a semi-structured interview designed to explore, in greater depth, their expectations regarding both the certification process and the post-certification monitoring process. This also included discussions around potential changes in these processes for high-risk systems, as well as participants' perceptions of fraud in AI certification and how it might be prevented. We chose to include drawing exercises because they can provide rich insights into how individuals conceptualize complex processes [35]. Semi-structured interviews were used for their balance of structure and flexibility, allowing for a deeper understanding of participants' perceptions [48].

3.1 Participants and Recruitment

We recruited participants for our study through a variety of channels, including mailing lists, flyers, poster advertisements, social media platforms, and word-of-mouth outreach. All participants were required to be at least 18 years of age.

The study included two groups: non-experts and AI experts. In the non-expert group, four participants identified as men and eleven as women, with an average age of 39.53 years ($SD = 19.14$, $Min = 21$, $Max = 70$). Participants had diverse educational backgrounds: eleven participants held a high school diploma, one had a bachelor's degree, and three held a master's degree. Regarding employment status, six were employed full-time, seven were students, and two participants were retired. A summary of demographic information for the non-expert group is provided in Table 1.

In the expert group, eleven participants identified as men and four as women. The average age in this group was 33.67 years ($SD = 5.35$, $Min = 26$, $Max = 43$). In terms of educational level, six participants held a master's degree and nine held a doctoral degree. The experts in our study come from diverse areas, including reinforcement learning, information security and privacy in AI, explainable AI, natural language processing, AI safety and trust, human-AI interaction, and responsible AI for public health applications. Demographic details for the expert group are summarized in Table 2.

3.2 Study Procedure

Each interview session with a participant consisted of five main parts. All questions asked during the sessions are provided on OSF². The sessions were audio-recorded, and the drawing process was also video-recorded. Each session lasted approximately one hour, with experts averaging 48.83 minutes ($Min = 30.34$, $Max = 67.29$) and non-experts averaging 33.24 minutes ($Min = 20.45$, $Max = 52.58$). Interviews were conducted separately for each group by a designated researcher: the first author conducted the interviews with experts, while the second author conducted those with non-experts. The detailed procedure of the study is as follows:

1) Consent & Demographics. First, participants were informed about the data being collected and their rights, including the ability to withdraw from the study at any time without any negative consequences. They were also told that the interview would be audio-recorded and that the drawing exercise would be video-recorded without capturing their faces. This information, along with details about data and privacy protection, was provided in an information sheet that included a consent form. Participants were asked to read the consent form and, if they agreed to its terms, to sign it. Afterward, each participant provided demographic information, including age, gender, education, and occupation.

2) Orientation and Warm-Up. To begin, we asked participants a few questions to understand their perspectives on AI systems, encouraging them to answer as openly as possible. We emphasized that the goal of the study was not to judge or test them, but to hear their honest thoughts.

For non-expert participants, we started by exploring their familiarity with the term "Artificial Intelligence" and asked what it

¹<https://osf.io/7j6vb/overview>

²<https://osf.io/h95k2/files/d4xb5>

ID	Age	Gender	Education	Job
P_N1	43	F	School diploma	Social Assistant
P_N2	70	F	Master	Retired (former teacher)
P_N3	67	F	Master	Retired (home economist, manager, Dipl. Social Sciences)
P_N4	69	F	Master	Dipl. Social pedagogue
P_N5	63	M	School diploma	IT systems administrator
P_N6	58	F	School diploma	Industrial clerk
P_N7	24	M	School diploma	Student
P_N8	29	F	School diploma	Student
P_N9	28	M	Bachelor	Government employee
P_N10	23	F	School diploma	Student
P_N11	21	F	School diploma	Student
P_N12	21	F	School diploma	Student
P_N13	22	F	School diploma	Student
P_N14	27	M	School diploma	Student
P_N15	28	F	School diploma	Secretary

Table 1: Demographic Information of Participants in the Group of Non-Experts.

ID	Age	Gender	Education	Job
P_E1	26	M	Master	PhD Candidate; Reinforcement Learning
P_E2	34	M	PhD	Postdoctoral Researcher; Information Security and Privacy
P_E3	37	M	PhD	Research Group Leader; Human-Centred Security
P_E4	30	M	Master	PhD Candidate / Scientific Researcher; AI in Legal Domain
P_E5	43	M	PhD	Research Group Leader; Human-Centric Explainable AI
P_E6	32	F	Master	PhD Candidate; Human-Centric Explainable AI
P_E7	27	M	PhD	Human-AI Interaction Researcher / ML Engineer
P_E8	28	F	Master	PhD Candidate; Uncertainty Quantification
P_E9	39	M	PhD	Professor; Business Analytics and Cyber-Physical Systems
P_E10	40	M	PhD	Senior AI Expert; Applied AI and Software Engineering
P_E11	27	M	Master	PhD Candidate; Information Systems
P_E12	41	F	Master	PhD Candidate; Responsible AI, Fairness in ML for Public Health
P_E13	34	F	PhD	Postdoctoral Researcher; NLP and LLMs
P_E14	31	M	PhD	Postdoctoral Researcher; AI Safety and Trust
P_E15	36	M	PhD	Professor; Human-AI Interaction, Explainability, Trust, and Human Oversight

Table 2: Demographic Information of Participants in the Group of Experts.

meant to them. They were also asked to name any AI-based systems they knew. To ensure a clear and common understanding, we provided a simple explanation of AI systems as "software that generates predictions, recommendations, or decisions using techniques like machine learning and logic-based methods. Examples include chatbots like ChatGPT, facial recognition for identity verification, autonomous vehicles, fraud detection in banking, and AI-powered medical diagnostics". After providing this definition, we asked again about their experience using AI systems, including which ones they had used, how often, and their comfort level with these technologies.

For expert participants, given their existing knowledge, we directly inquired about their definition of AI, the AI systems they use in their work or daily lives, and their feelings about these systems.

Next, for both groups, we discussed trust and certification labels. Participants were asked what factors would make them trust an

AI system and what they understood by the term "certification label" in this context. To ensure consistency across participants, we provided the following definition of a certification label: "It is a mark or symbol applied to a product, service, or system to indicate that it meets certain predefined standards, regulations, or criteria. These labels are typically issued following an assessment or verification process. They serve as a guarantee to consumers or users that the product or service has been tested, evaluated, and verified for compliance with established benchmarks". After that, we asked participants about their views on AI certification and how certification labels on AI systems affect their trust in those systems, and examples of situations where AI certification might be especially important.

3) Drawing Exercise. Following the orientation, participants were asked to engage in a drawing exercise to illustrate their expectations of the certification process for AI systems. We explained

that, similar to products like home appliances that require safety certifications, AI systems must also undergo a process to receive certification labels. We asked participants to depict the steps they believed an AI system should go through to be certified. After completing their drawings, we asked follow-up questions to clarify and elaborate on the elements and flow illustrated in their sketches.

To assist with the drawing task, we provided participants with a variety of printed icons representing elements such as an individual, a group of individuals, an institution or company, a server, a document, a checklist, a smartphone, the internet, data processing, monitoring, and a certificate. These icons were intended to make the drawing process easier, and participants were encouraged to use any that were helpful for their sketch. However, they were also free to draw entirely on their own if they preferred. During this exercise, participants were encouraged to think aloud [11], explaining their choices and what each part of their sketch represented. This approach allowed us to better understand their thought processes and expectations related to the AI certification process.

4) Semi-Structured Interview. We conducted semi-structured interviews to further explore participants' perceptions and expectations. We refined the interview script through pilot interviews that allowed us to identify and address potential issues with question clarity, structural flow, timing, and participant comfort. During these sessions, researchers took notes and documented participant feedback, which was then used to make improvements to the questions and interview procedure.

In the interview, we examined participants' views on which entities should issue certificates and their trust in certifications issued by government agencies compared with those issued by NGOs or private companies. We also explored the desired level of detail about the certification to accompany the label and whether they preferred the information to be tailored to users' varying levels of technical expertise. Additionally, we investigated post-certification monitoring, focusing on its perceived importance, responsible bodies, verification frequency, consequences for non-compliance, and desired transparency levels. Further, we examined whether the certification process or post-certification monitoring should be modified for high-risk AI systems. Finally, we explored fraud in AI system certification, discussing participants' definitions of fake labels, related concerns, perceived protection measures, and when and how they would verify the authenticity of certification labels.

5) End & Reimbursement. At the end of the interview, participants were invited to ask questions and share any additional feedback. Finally, we provided each participant with 15 euros as compensation.

3.3 Ethical Considerations

In conducting our study, we adhered to the guidelines set by the ethics committees in the institutions of the authors. At our institutions, user studies must restrict the gathering of personal data to safeguard the participants' privacy. Every participant was given a random identifier. Before the interview, each participant signed a consent form, which was stored separately from all other information to ensure that it could not be linked to their identities. Prior to the study, we obtained approval from the ERB at the University of Duisburg-Essen.

3.4 Data Analysis

To maintain the privacy of participants, all collected data were anonymized prior to analysis. Audio recordings were transcribed, and personal information was replaced with neutral markers. To prevent identification through handwriting in the drawings, machine-generated text was placed over the handwritten text using a picture editing tool to conceal it. The sketches and interview transcripts were then analyzed separately using thematic analysis [12].

First, we analyzed participants' sketches to capture their perceptions and expectations of the certification process. To do this, we used an open-coding approach with two researchers (the first and second authors) serving as coders. By reviewing all sketches from both groups, the two coders generated and agreed on a final codebook. The codebook included three main codes: type of certification process flow (sequential or interactive), stakeholders and process elements, and controlling entities (i.e., decision-makers involved throughout the certification process). Then, each coder coded the sketches of one group (experts or non-experts), while the other coder acted as a reviewer for that analysis. The results were discussed, and final code allocations for each drawing were decided collaboratively. Throughout the analysis, we also considered the audio recordings to supplement information expressed in the sketches when parts of the drawings were unclear.

Second, we analyzed the interviews using a hybrid coding approach: we started with a small set of deductive codes derived from the semi-structured interview guide and remained open to emerging themes during analysis, adding inductive codes to capture concepts relevant to our research questions. For each participant group, one researcher who conducted the interviews and was familiar with the data (the first author for the experts group and the second author for the non-experts group) developed an initial codebook. The two researchers then met to compare and discuss their codebooks, agreeing on a single final version. This final codebook also included some codes unique to either the experts or the non-experts group. It comprises 5 categories and 98 codes (the complete codebook is available on OSF³). Using this final codebook, the interviews were coded following established thematic analysis guidelines [12, 21]. The first author coded the expert interviews, with the second author reviewing and verifying the coding. Conversely, the second author coded the non-expert interviews, while the first author served as reviewer and verifier. Finally, both researchers met to discuss and resolve any disagreements.

4 Results: Experts vs. Non-Experts

This section presents the results of interviews conducted with both non-experts and experts, as well as the outcomes of their sketches. The findings from our thematic analysis are organized by theme. To indicate how frequently certain aspects were mentioned, we include counts of participants who mentioned each aspect; however, these counts are intended only to illustrate trends and do not constitute a quantitative analysis of the data. We refer to non-experts as P_N (N_{nonexp} for descriptive counts) and experts as P_E (N_{exp} for descriptive counts).

³<https://osf.io/h95k2/files/m3sx8>

4.1 AI Knowledge, Experience & Trust

As described in Section 3, we first asked participants in both groups orientation questions about how they define, use, and trust AI. Although the questions do not directly address the study's main research questions, they provide important context for understanding participants' prior knowledge and experiences, which helps interpret their perspectives on AI certification. We therefore report here the results of these orientation questions.

Different Views on the Meaning of AI. Participants described AI in diverse ways, with experts, as expected, providing more detailed and technical descriptions. Table 3 summarizes the perspectives of participants in the two groups. Many participants, both experts and non-experts, described AI as learning models or systems that adapt and improve from data. Experts often mentioned large language models (LLMs) as examples, while non-experts described AI more generally as programs capable of learning. Another common understanding was AI as tools for assisting with tasks or decision-making. Experts emphasized AI as systems that help explore data, build models, and support decisions, whereas non-experts saw it as technology that helps people in everyday tasks or decisions. Some participants associated AI with human-like behavior or capabilities. Experts highlighted AI's ability to perform tasks that appear intelligent or mimic human reasoning, while non-experts described it as technology that acts or thinks similarly to humans. Non-experts also emphasized AI as a tool for data classification and retrieval, describing systems that organize, gather, and provide information based on queries. This perspective was not mentioned by experts. Finally, a few experts characterized AI as a "moving target", reflecting its evolving definition over time.

Frequent AI Use, Experts Cautious, Non-Experts Optimistic. Most participants, both experts and non-experts, reported frequent use of AI systems in daily life ($N_{\text{exp}} = 12$, $N_{\text{nonexp}} = 10$), with generative AI (GenAI) tools being the most widely used ($N_{\text{exp}} = 14$, $N_{\text{nonexp}} = 13$). Indirect AI use via software with embedded AI features, such as recommendation functions, was also recognized ($N_{\text{exp}} = 5$, $N_{\text{nonexp}} = 9$). Also, a few mentioned the usage of AI-based software development tools ($N_{\text{exp}} = 2$, $N_{\text{nonexp}} = 1$). Despite frequent usage, participants expressed different feelings toward AI. Among non-experts, most expressed positive feelings ($N_{\text{nonexp}} = 13$). On the contrary, most experts reported mixed feelings ($N_{\text{exp}} = 9$), meaning they experienced both positive and negative reactions or uncertainty, while some expressed only positive ($N_{\text{exp}} = 3$) or only negative/concerned ($N_{\text{exp}} = 3$) views. For example, P_{E12} said, "I know how they work, how the underlying systems work. I do have concerns, for example, about privacy and fairness".

Shared Views on What Builds Trust in AI, with Trust Barriers Highlighted Only by Experts. Trust in AI systems was linked to several factors. Adherence to well-established regulations and good standards was frequently mentioned by both experts and non-experts ($N_{\text{exp}} = 11$, $N_{\text{nonexp}} = 8$). Non-experts associated trust in AI systems particularly with adherence to transparency. For instance, P_{N9} stated, "Definitely transparency about the decision-making process behind the answer it gives me".

Reliability, accuracy, and utility were other common factors. Experts highlighted these aspects ($N_{\text{exp}} = 7$), with P_{E15} commenting,

"I do think it has a lot to do with your personal experiences and the benefits you see in using AI. So if you see a lot of benefits, if you have experienced that it works for some tasks, that makes you trust that. So that's the functionality aspect of a trustworthy AI". Non-experts also emphasized reliability and good performance ($N_{\text{nonexp}} = 3$), with P_N10 noting that consistent performance and useful results enhance trust.

Experts also expressed trust in systems coming from reputable providers or evaluated and certified by trustworthy institutions ($N_{\text{exp}} = 4$), or recommended by a colleague ($N_{\text{exp}} = 1$). Non-experts did not report similar considerations, focusing more on system behavior than provenance.

Factors undermining trust were mostly reported by experts. These factors include risks associated with AI systems ($N_{\text{exp}} = 5$), unreliable outputs ($N_{\text{exp}} = 1$), and untrustworthy service providers ($N_{\text{exp}} = 1$). Non-experts did not identify negative factors, with only one participant citing a lack of transparency. This could be likely due to limited technical knowledge and the generally positive attitudes toward AI expressed by the group.

4.2 Perceptions & Expectations of AI Certification

In this section, we report participants' perceptions and needs regarding AI certification, including the impact of systems being certified on their trust in these systems, who should be responsible for certifying AI systems, and their requirements regarding transparency in AI certification (i.e., what information they want and at what level of detail). We also present the results of sketches drawn by participants, illustrating how they envisioned the certification process and its different stages. The findings presented in this section help address RQ1 (see Section 1).

Certification Increases Trust Among Non-Experts More than Experts. We asked experts and non-experts about the impact of AI systems having certification labels on their trust in these systems, and we found that the perceived effect varied between the two groups. Non-experts were generally more positive about certification, with most participants ($N_{\text{nonexp}} = 12$) reporting that it would increase their trust. For instance, P_N14 explained, "Well, certification would definitely increase trust. It would make it clearer that the AI can be trusted, and this trust would likely be reflected in the quality of the data it provides compared to a non-certified AI", and P_N4 added, "I would then assume that the content I share there would be handled more safely". In contrast, few experts ($N_{\text{exp}} = 3$) reported a positive impact of AI systems being certified and displaying certification labels. Four experts and one non-expert expressed mixed or conditional effects. For example, P_E12 described certification as "a good step in the right direction, but it's not the fix-it-all solution", while P_E2 noted, "The impact on trust could go both ways. Some certifications could increase the trust I have in an AI-based system, but they could also decrease it depending on the entity providing the certificate". Experts were also more negative about the impact of certification than non-experts ($N_{\text{exp}} = 3$, $N_{\text{nonexp}} = 1$). For instance, P_E13 noted, "I don't think any labeling affects my usage patterns or my trust in AI", and P_E7 warned, "Providing a certification could have adverse effects because people tend to over-rely on AI... if you

Definition Aspect	N_{exp}	N_{nonexp}	Example Quote
AI as learning models	10	4	P_{E11} : "AI is anything that learns from data...often instantiated nowadays in LLMs"; P_{N11} : "It's a program that is capable of learning"
AI as assistance for tasks or decision-making	5	6	P_{N15} : "For me, artificial intelligence actually means support, at least in everyday life"; P_{E12} : "AI is a set of tools that help us explore data, understand it, and create models to make decisions within a particular context"
AI as human-like behavior or capabilities	3	3	P_{E1} : "AI systems are those that seem human-like to humans"; P_{E9} : "AI is the case when the computer shows signs of human intelligence...it always consists of sensing, processing, and acting"; P_{N5} : "Kind of like a super brain or something that can answer any question"
AI as data classification and retrieval	0	4	P_{N4} : "It's about gathering all available data from various areas, categorizing it using regular computer systems, and then delivering it based on the query"; P_{N8} : "Artificial intelligence, by that I mean a system that collects data... and then provides answers accordingly"
AI as a "moving target" or evolving definition	2	0	P_{E15} : "AI is often what people associate with the coolest stuff currently out there...what was considered AI 10 years ago may not be considered AI anymore"

Table 3: Expert and Non-Expert Perspectives on What AI Means.

put a sticker on it that says, 'hey, this is trustworthy,' I think this only makes it worse".

Experts and non-experts identified specific situations in which AI certification can be most useful or likely to promote trust. Both groups emphasized high-risk AI systems as a key context ($N_{\text{exp}} = 6$, $N_{\text{nonexp}} = 6$). Experts additionally highlighted the importance of certification when the certifier or regulator is trustworthy ($N_{\text{exp}} = 5$) and when the certification process is robust and adheres to good standards or meets certain conditions ($N_{\text{exp}} = 9$). For instance, P_{E8} said, "Certification labels would be helpful only if they are provided for AI systems based on models that can be robustly verified", and P_{E7} stated, "Certification labels would make sense only if certification was conducted for certain aspects, as it is difficult to certify all aspects of an AI system". Two experts noted that certification would be particularly beneficial for lay users. Furthermore, non-experts ($N_{\text{nonexp}} = 4$) mentioned that certification could be particularly useful for widely used AI systems. For example, P_{N7} stated that certification would be particularly important for free, easily accessible tools like ChatGPT, or Canva AI, as "that's what most people use".

Certificates to Be Issued by Independent Agencies; Experts More Open to Private Companies. We inquired about experts' and non-experts' preferences regarding which entities should be responsible for issuing AI certification labels. Figure 1 summarizes the reported preferences. Note that some participants indicated more than one preference; for example, certification could be carried out by either government agencies or NGOs.

As shown in Figure 1, independent organizations or NGOs were the most frequently cited choice. The reasons behind this preference were mainly to ensure neutrality and unbiased oversight, as well as independence from political or financial influence. P_{N6} explained, "Definitely independent institutes—ones that are not backed by financial backers from the business world or any political parties, but are completely independent and therefore completely free to make

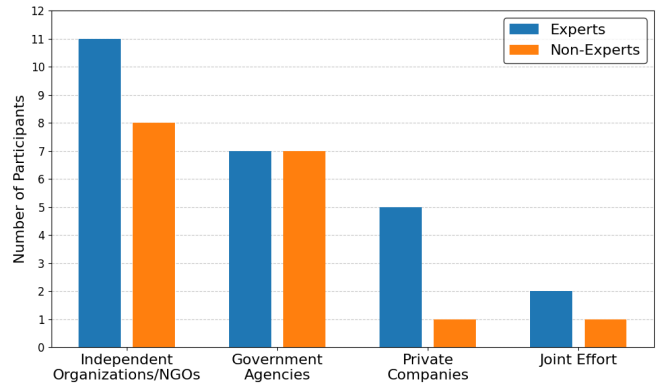


Figure 1: Who Should Certify AI Systems?

their own decisions". Similarly, P_{N15} added, "I think I would trust the nonprofit more because I would feel that they would not benefit from it and would not manipulate the data in any way."

Government agencies were the second most common choice in both groups. For example, P_{E2} stated, "As I live in a democratic country and I tend to trust the system, so I would more trust a governmental institution". Also, P_{N7} mentioned, "I think I would trust the state-certified one. Because it gives me the impression that it definitely involves laws, a lot of discussion and, above all, possibly also legal decisions". P_{N14} added, "I would rather trust the government certification because, for example, if you look at the EU law, you can see that it tends to protect the user or consumer rather than associations or companies making the rules themselves".

A smaller number of participants proposed that issuing certificates could be a joint effort among multiple institutions, such as government bodies and NGOs collaborating on the process. Experts

also mentioned that private companies could issue certificates, often motivated by perceived expertise gaps, whereas only one non-expert supported certification by companies. P_{E8} stated: *"At the moment, I don't think the government has actual AI experts. The field is too new and the government doesn't pay as much as the industry. So the best people can be found in the industry"*.

Some concerns regarding certification by either private companies or governmental institutions were also raised by experts. Regarding private certification, P_{E14} said, *"if a company develops the system and that company also certifies it, then... that's a very typical case of conflict of interest"*. Similarly, P_{E1} expressed skepticism toward private certificates, viewing them as potentially serving as a *"marketing thing"*. Concerns about governmental certification centered on the risk of political influence.

Transparency Has a Greater Positive Impact on Non-Experts' Trust than on Experts. Both experts and non-experts highlighted the importance of providing clear and relevant information alongside AI certification labels. Most participants ($N_{\text{exp}} = 8$, $N_{\text{nonexp}} = 8$) wanted details about the certification process, with experts in particular interested in knowing what was certified and how it was certified. Participants also emphasized the importance of information about the certification authority ($N_{\text{exp}} = 4$, $N_{\text{nonexp}} = 5$). Only experts ($N_{\text{exp}} = 2$) valued details such as the certificate's issue and expiration dates, the specific system version certified, and the potential harms that may persist despite certification. Non-experts, on the other hand, prioritized information on how data and privacy are handled in certified systems ($N_{\text{nonexp}} = 4$).

Regarding the level of detail needed, seven experts and several non-experts considered providing only detailed information acceptable, while nine non-experts and only one expert felt that providing only basic information would still be sufficient. However, the majority of participants preferred offering both basic and detailed information ($N_{\text{exp}} = 12$, $N_{\text{nonexp}} = 10$) to accommodate users with varying levels of expertise. They emphasized that technically knowledgeable users would benefit from highly detailed information, whereas lay users would find simplified explanations more useful. As P_{E11} noted, *"Different users need different types of granularity regarding the information load that they receive"*.

The perceived impact of providing additional information alongside certification labels varied between groups. Most non-experts ($N_{\text{nonexp}} = 12$) were positive, reporting that such information would increase trust. In contrast, experts expressed less positive views: four warned that combining certification labels with detailed information could overwhelm lay users and doubted whether lay users would invest the time to understand complex certification details (P_{E8} : *"overloading people with information and therefore leading to less trust, confusion, whatever"*), two suggested it would have no impact on their trust, and only four believed it would promote their trust (P_{E8} : *"If you explain to me how it's built and how it was certified, of course, I would be able to calibrate my trust"*).

Non-Experts Viewed Certification Process as Sequential, Experts as Interactive. From the sketches, we identified two types of AI certification processes imagined by participants: sequential and interactive. Sequential processes follow a step-by-step order, with each stage building on the previous one and ending with the issuance of a certificate. Interactive processes, on the other hand,

involve collaboration, feedback, and active interaction among stakeholders, rather than following a straight line.

The sequential approach was strongly preferred by non-experts ($N_{\text{nonexp}} = 13$) compared to experts, as only six experts sketched a sequential process. However, it is worth noting that the sequential process sketches from non-experts were generally more basic and less detailed than those of experts. Most non-experts depicted a simple process that begins with AI development, proceeds to authority evaluation, and concludes with the issuance of an AI seal to end users. Figure 2b shows an example of a sketch illustrating a more detailed sequential AI certification process drawn by a non-expert. Specifically, P_{N8} 's sketch depicts a linear pathway in which AI software is first tested internally by employees, then evaluated on test subjects, reviewed by a governmental agency and experts, and finally retested to ensure it is error-free. Once all requirements are met, the process concludes with the issuance of an AI seal. Figure 2a illustrates an example of a sequential certification process drawn by expert P_{E11} . In this process, a company develops an AI model that undergoes governmental checks based on several criteria, including infrastructure, ethics, fairness, and data privacy and security. If the model passes evaluation, it receives certification and can be deployed to end users. For high-risk applications, certification is mandatory and managed by a specialized agency or subdivision to ensure stricter standards.

Interactive certification was envisioned more strongly by experts ($N_{\text{exp}} = 9$), who emphasized multi-stakeholder collaboration and distributed decision-making in both the design and implementation of certification. Figure 2c illustrates an example of an interactive certification process from expert P_{E15} . In this model, five key stakeholders interact within a feedback-rich system: a developer creates an AI system, which must be tested by a certification body against standards developed by standardization institutes (e.g., ISO) under government mandate, before being deployed by companies or end users. The interactive elements include bidirectional feedback loops between all parties. For example, standards bodies can revise criteria based on certification challenges; failed systems are returned to developers with targeted improvement requirements; deployers report real-world issues back to certification bodies; end users flag problems to deployers; and certification bodies provide implementation feedback to regulators. Figure 2d presents an interactive certification process sketched by non-expert P_{N9} . As explained by P_{N9} , data about the AI system is provided to the certification authority for evaluation. During the verification phase, the submitted information is checked against defined standards. This phase is iterative: the certification authority may request corrections or additional information, which the system owners must provide, and the process is repeated until compliance is achieved. Once the verification phase is complete, the system can be published online with a certification seal indicating official approval.

Analysis of sketches also revealed four fundamental stakeholders in AI certification: certification agencies, AI developers, end users, and government regulators. All experts granted primary controlling power to certification agencies, with many extending influence to users and regulators, particularly in shaping certification metrics and standards. Several experts stressed the need for public and user involvement in certification processes ($N_{\text{exp}} = 5$). For example, P_{E7} recommended mechanisms that *"allow users to report like dangerous*

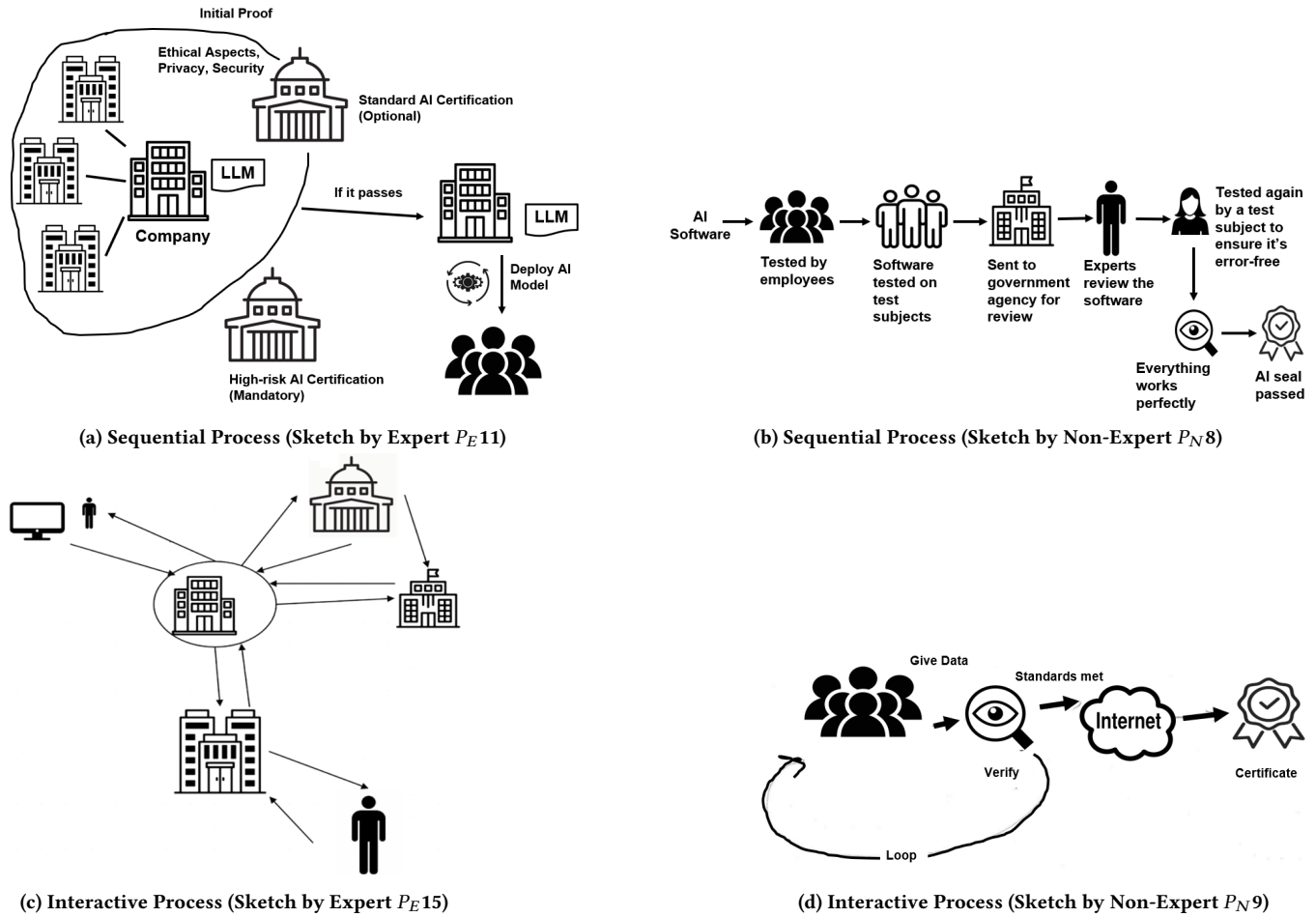


Figure 2: Examples of AI Certification Processes from Experts and Non-Experts.

or unwanted outputs, outcomes, so that they can aid the certification process".

Strong Emphasis on Laws, Standards, and Metrics. Almost all participants emphasized the need for comprehensive laws, well-defined standards, and clear evaluation metrics in order to achieve effective certification.

Non-experts primarily emphasized the need for ethical and legal considerations and raised concerns about data management and information security. They stressed the importance of protecting user data as a prerequisite for certification, though their insights lacked the depth provided by the experts.

Experts shared many detailed insights and offered various suggestions. For example, P_{E4} and P_{E14} noted that one of the certification standards should require multiple evaluators to ensure objectivity and reliability. Specifically, P_{E4} stated: "it must be tested at least by two different labs and get similar results...And these labs must be anonymous to each other to prevent conflicts of interest and ensure discrete, confidential evaluation processes". Further, P_{E7} highlighted the need for specialized evaluation standards for AI systems used in decision-support roles, such as in the medical domain, where AI

complements rather than replaces human physicians. According to P_{E7} , in such cases, the standards evaluation and certification process should assess the performance of the human-AI team as a whole, rather than focusing solely on the AI system.

Moreover, many experts ($N_{\text{exp}} = 7$) highlighted that certification must be context-specific and tailored to different use cases, as AI systems are too diverse and complex for a one-size-fits-all approach. For example, P_{E7} stated: "I think..the trustworthiness and security of an AI system, it's more complex than just having this binary yes or no label. I think you can certify the system for certain aspects, and maybe you can identify regions of the input space where the system is less trustworthy. But ultimately, I think it's input and context dependent and not as simple as saying the entire system is trustworthy or not". They also stressed that evaluation should extend beyond the AI model itself to include the data used for training and the infrastructure supporting the system. Additionally, experts (P_{E14} and P_{E12}) suggested that certification results should be communicated using a clear, intuitive scale or performance level—similar to grading systems or labels like gold, silver, and bronze—so that all stakeholders, especially lay users, can easily understand them.

4.3 Perceptions & Expectations of Post-Certification Monitoring

Since AI systems can evolve significantly over time, post-certification monitoring can be crucial to ensure continued compliance with certification standards. Therefore, we asked participants about how important they consider monitoring AI systems after certification. The majority, especially experts, recognized this importance. This section presents our findings on participants' views and requirements regarding post-certification monitoring, including how frequently monitoring should occur, how it should be conducted, by whom, and what consequences should follow if a certified AI system fails to meet certification standards over time. The results of this section also contribute to answering RQ1.

Monitoring Frequency: Experts Favor Update-Based Monitoring, Non-Experts Favor Annual Checks. We asked both experts and non-experts about the frequency of post-certification monitoring for AI systems, and they provided various opinions. The most frequently mentioned trigger by experts was with every new release, feature, or update ($N_{\text{exp}} = 7$). For example, P_{E11} explained, "AI models can change much quicker than a car...So you technically should need a new certification every time the model updates itself". Non-experts also mentioned this trigger, but less frequently ($N_{\text{nonexp}} = 3$), with P_{N14} stating, "You would have to look more at the versions, when major leaps take place... there will be many AI leaps in a very short time when it comes to data ...".

Yearly monitoring was proposed by a smaller group of experts ($N_{\text{exp}} = 2$) as a manageable interval. P_{E5} stated, "every year...maybe you get a first label and then every year...you go through a process just to kind of verify that nothing has changed". Among non-experts, yearly or every few years was the most common suggestion ($N_{\text{nonexp}} = 7$), e.g., P_{N9} said, "It should be reviewed every one to two years".

Experts additionally highlighted that monitoring frequency should vary based on the system's characteristics and available resources ($N_{\text{exp}} = 4$). For instance, P_{E15} said, "This probably depends a lot on the system...you need to define what the monitoring interval is and what exactly to monitor. And the monitoring interval can depend on updates, can depend on changes in the environment, and changes in who's working with the system". Non-experts implicitly considered sensitivity and usage context when suggesting shorter intervals (monthly or every few months, $N_{\text{nonexp}} = 6$), especially for high-impact applications. P_{N10} emphasized, "Annually would be too long... I would perhaps say every two or three months".

Finally, participants from both groups acknowledged uncertainty regarding optimal frequency. Experts ($N_{\text{exp}} = 2$) openly questioned feasibility and practical implementation (P_{E13} : "I am not sure how we can deploy it actually in real life because nearly every month GPT introduces a new model..."), whereas non-experts often framed uncertainty in terms of unfamiliarity with AI technology (P_{N4} : "I don't know how quickly things change in this field... maybe once a year? But I'm not sure if that's enough").

Post-Certification Monitoring Process: Experts Push for Stricter Rules, Non-Experts for Transparency. Experts and non-experts highlighted some expectations for the post-certification

monitoring process. Stricter regulations to ensure AI systems continue following certification metrics were advocated by 4 experts, with P_{E1} explaining, "companies should be obliged to notify whoever is in charge of certification whenever they make an update". System history tracking (i.e., keeping a record of the system's changes over time) was highlighted by 2 experts as important for maintaining trust and transparency. Further, the need for a fully automated post-certification monitoring process to handle the speed of AI development was mentioned by 2 experts, with P_{E11} suggesting, "this would require a fully automatized pipeline where they just send in the new version of the LLM...and it goes through automatized standardized tests within the AI certification agency and they automatically get back the new certification". Monitoring when systems are re-trained on new data was emphasized by 3 experts, as P_{E14} noted, "if the audit has been performed on a specific dataset on which the AI model was trained, then it is important to ensure that any additional data which does add up...then we need to do an audit once more".

Non-experts also emphasized transparency ($N_{\text{nonexp}} = 12$), focusing on clear, accessible documentation of updates in development, and performance. For example, P_{N6} stressed, "who is monitoring and certifying, and perhaps what criteria have been used for verification... how secure the whole thing is". Participants also highlighted the need for user-friendly formats, with P_{N14} noting, "it would be very useful to see who is doing the verification and what the verification process steps are, explained in broad terms, perhaps with some visual aids... I would take a look at something like that, but I don't want to go into a report and dig around to find out how these monitoring processes take place".

Experts and non-experts expressed differing views on whether the same entity that issues AI certification should also conduct post-certification monitoring. Among experts, they favored the same entity handling both tasks, mentioning simplicity and efficiency. For instance, P_{E2} stated, "having additional institutions here would mostly add too much complexity". Only one expert suggested that different entities could be beneficial or that parallel monitoring by independent agencies could increase trust, as P_{E3} noted, "do like parallel certification by independent agencies...hopefully they come to similar results. This redundancy can also increase the trust". Non-experts were divided on this issue. Seven participants supported the same entity handling both certification and monitoring. Another seven participants preferred separating these responsibilities between different entities, and that is to ensure independence and reduce potential bias. For example, P_{N12} explained, "Maybe companies could do that too—because if the certification itself was issued by a governmental body, you could assume that the standards are high. And then it's just a matter of checking whether they are still being upheld. Even with small changes, a company could step in". P_{N13} added, "A new organization should be created for this purpose".

Post-Certification Non-Compliance: Certificate Withdrawal, System Shutdown, and Opportunities to Fix. We asked experts and non-experts about what should happen when AI systems fail to meet certification standards, and their perspectives were overlapping. Most non-experts suggested certificate withdrawal as a consequence ($N_{\text{nonexp}} = 8$), with P_{N4} noting, "Then the label should be removed". Certificate withdrawal as a formal consequence was also discussed by experts ($N_{\text{exp}} = 4$), with P_{E5} comparing it to

restaurant health inspections: *"if they don't meet the standards...they revoke the license until they can show that they have achieved the standards"*.

Additionally, system shutdown or withdrawal was recommended by participants ($N_{\text{exp}} = 4$, $N_{\text{nonexp}} = 4$), especially for high-risk cases. For example, P_{N1} said: *"Then I would say the product should be taken off the market"*, and P_{E15} noted, *"I would say we can't use it anymore...depending on the risk, you should then also say, hey, we cannot use that now anymore until we have fixed it"*.

Financial penalties were suggested by experts ($N_{\text{exp}} = 3$). Moreover, many participants ($N_{\text{exp}} = 6$, $N_{\text{nonexp}} = 4$) emphasized giving companies a chance to fix issues before severe penalties. P_{E12} stated, *"they could and should have a second chance and try to get re-certified"*. Similarly, P_{E13} said, *"I think they should get a chance to fix it. Maybe firstly, they need to be warned about it"*, and P_{N5} mentioned, *"The provider should be informed and asked to fix it. If not, the certification should be revoked."*

Some experts suggested making people aware of the non-compliance ($N_{\text{exp}} = 4$), rather than shutting down systems. For example, P_{E11} explained, *"I don't think that it's practical to shut down AI models because that would just cause a bunch of damage...there should be notifications of...here's a problem. The AI system has failed the certification test. This is why it failed the certification test"*. P_{E15} added, *"we should tell people this is not certified anymore...and then have a red warning signal or whatever to make people really aware of what is happening here"*. Two non-experts also highlighted public awareness, with P_{N11} noting, *"They... issue reprimands... and make them public. I think it makes the most sense... so that when a certification expires, you also inform people about it"*.

4.4 How Certification Should Change for High-Risk AI

We asked participants about potential changes in certification processes for high-risk AI systems (this addresses RQ2). The most frequently mentioned modification was the implementation of stricter regulations, standards, and evaluation procedures ($N_{\text{exp}} = 12$, $N_{\text{nonexp}} = 13$). Experts emphasized comprehensive evaluations, e.g., P_{E12} said: *"There should be additional tests in real-world settings or scenarios where there is a period of time...where this system is still not fully in production and is being tested to see exactly how it behaves in a real-life setting"*. Also, mandatory certification for high-risk applications was mentioned, P_{E11} noted: *"I think for high-risk applications, it needs to be mandatory to pass certification in order to deploy it"*. One expert also highlighted the importance of context-specific evaluation approaches. P_{E7} explained, *"Autonomous driving is an example of a fully autonomous system...when you have a decision support system, you should really evaluate the decision support and the decision maker as a team"*. Non-experts focused more on strict oversight, frequent verification, and transparency. For example, P_{N14} mentioned: *"There should be really high standards...the government...should be more responsive to users or consumers"*, and P_{N15} stated: *"Yes, regularly and very precisely, that it's handled very sensitively, that everything is always up to date and accurate"*. Additionally, P_{N8} said: *"Additional information should be provided...if you were to use it in healthcare, people's lives would be at risk"*; P_{N7} :

"There needs to be more data, more knowledge, and above all more research".

A smaller number of participants in both groups suggested assigning different responsible entities for high-risk systems ($N_{\text{exp}} = 2$, $N_{\text{nonexp}} = 2$). For instance, P_{E14} proposed hierarchical or organizational adjustments: *"There needs to be an international body that can say the system is trustworthy, then it needs to be voted as trustworthy by X number of other bodies"*, whereas P_{N4} suggested: *"Could there be something like the four-eyes principle to double-check decisions"*.

4.5 AI Certification Fraud and Prevention

Ensuring the integrity of AI certification is critical for maintaining trust in certified AI systems; thus, we asked participants about how they perceive the risks of AI certification fraud and the prevention strategies they consider effective to maintain the credibility of AI certificates. This section presents our findings, which address RQ3.

Types of Fraudulent AI Certifications: Copying, Self-Issued, Unauthorized. Both experts and non-experts identified multiple types of potential fraudulent AI certification labels. These types are summarized in Table 4. The most frequently discussed forms include copying real labels without certification, obtaining labels without proper evaluation, self-issued labels, and labels from unauthorized entities. Experts also highlighted sophisticated fraud, such as submitting one system for certification but deploying a different one. Additionally, experts expressed concerns about the difficulty for end users in distinguishing real from fake certifications and the higher risk of such fake AI certificates when issued by private companies. For example, P_{E2} explained, *"There is a higher risk of fake certifications if certificates are attributed by private entities... it may be more difficult to check if they actually do their job"*, while P_{E1} noted, *"A standard user would never do it. He would never... care to do it. That's a problem"*. Non-experts also mentioned similar concerns, although to a much lesser extent.

Technical, Legal, and Awareness-Based Countermeasures. Participants suggested multiple strategies to protect against fake AI certifications. Technical or security solutions were most frequently mentioned by experts ($N_{\text{exp}} = 9$), including digital labels with signatures or identification systems for AI models (P_{E9} : *"Those labels should be digital labels, which we can trust to contain a signature"*; P_{E11} : *"Having some sort of mechanism or ID for deployed LLMs, or maybe using distributed ledger technology—an identification system for LLMs"*). Only one non-expert mentioned a technical solution, highlighting the need for secure digital AI certification labels.

Strict laws, enforcement mechanisms, and penalties were emphasized by both groups; however, this was mentioned more by non-experts ($N_{\text{exp}} = 5$, $N_{\text{nonexp}} = 12$). For example, P_{N3} said: *"Strict laws, tight requirements, constant checks"*; P_{N1} stated: *"In my opinion, that's a criminal offense. So, there should be penalties"*.

Transparency and awareness-raising were mentioned by both experts ($N_{\text{exp}} = 5$) and non-experts ($N_{\text{nonexp}} = 3$), stressing the importance of educating users about certification processes and warning signs of fraud. Experts also suggested creating one or a few trusted labels to reduce confusion ($N_{\text{exp}} = 3$), a measure also

Type of Fraud	N_{exp}	N_{nonexp}	Example Quote
Using a genuine-looking certificate without undergoing certification	5	2	P_{E15} : "Somebody who's just copying an official certificate, putting it on top of their thing... there was no certification process in the background"
Paying for a certificate without completing the required evaluation	5	8	P_{N11} : "I would almost go so far as to say those that are simply paid for... that's lying about the trust that consumers have"
Labels created and applied by the system owner without independent oversight	5	1	P_{E4} : "I prepare a certificate for my own product. That is fake..."
Labels issued by organizations not authorized or recognized as legitimate certifiers	3	3	P_{N7} : "If some random company makes a label and says it's official, but it's not recognized anywhere, that's not trustworthy."
Certifying one system but deploying a different version or model in practice	1	0	P_{E11} : "They... send the wrong LLM to the agency. And then for deployment, they use a completely different LLM"

Table 4: Expert and Non-Expert Perspectives on Fraudulent AI Certification.

proposed by non-experts ($N_{\text{nonexp}} = 2$). Additionally, several participants in both groups recommended mechanisms for reporting or tracking fraud, involving authorities and the public ($N_{\text{exp}} = 2$, $N_{\text{nonexp}} = 2$), as well as public registries or information sources to verify certifications ($N_{\text{exp}} = 2$, $N_{\text{nonexp}} = 4$).

Verification of AI Certification Labels: Before Use or in Risky Situations. We asked participants about situations in which they would verify whether AI certification labels are real or fake. Checking labels before deciding to use a system was frequently mentioned by experts ($N_{\text{exp}} = 7$), with P_{E9} explaining, "I would first check it when I am interested in using the solution... I would check which of them have a certificate and which don't" and P_{E12} stating, "As a user, when would I check? Like the first time I use the system. And if it doesn't have a label, then I would think twice about using that system". Non-experts mentioned this trigger less frequently ($N_{\text{nonexp}} = 2$).

High-risk situations were highlighted by both groups, but they were more frequently mentioned by non-experts ($N_{\text{exp}} = 3$, $N_{\text{nonexp}} = 8$). For instance, P_{N6} mentioned: "Yes, in cases where something dangerous could actually happen..."; P_{N4} said: "Yes, if it really affected me. Like if it was a medical diagnosis relevant to me, or if the topic was important".

Verifying labels when the system or its output seems suspicious, or when a label appears uncommon, was another aspect mentioned by both groups ($N_{\text{exp}} = 2$, $N_{\text{nonexp}} = 3$). Participants indicated they would check labels under these circumstances. For example, P_{N5} noted: "Normally, I wouldn't. I'd trust the system. But if I got strange results—or if different AIs gave wildly different answers—I'd get suspicious"; P_{N10} stated: "Maybe if I notice that... strange or incorrect answers appear, or something doesn't seem right to me... then I would definitely question it and check it out".

5 Discussion

The findings from our study reveal significant differences between experts and non-experts in their perceptions, expectations, and approaches to AI certification.

Both user groups form trust in AI certification in distinct ways. Non-experts relied heavily on certification labels as cues for trust,

while expert users approached trust more cautiously and emphasized that their confidence depended on understanding the mechanisms behind the certification. These findings relate to the concept of trust calibration [73], as non-experts may overtrust based solely on interface cues, while experts tend to exhibit more calibrated trust. This pattern additionally aligns with HCI research showing that users who lack a clear understanding of AI systems often rely on intuitive heuristics, whereas those with deeper knowledge evaluate AI outputs critically and remain aware of potential limitations [53]. The differing attitudes of the two user groups may also explain mixed findings on certification's impact on trust. Some studies report increased trust [24, 56], while others find no significant effect [72]. Our results suggest that such inconsistencies may be partly due to variations in participants' technical expertise across studies.

Transparency was appreciated by both user groups, with non-experts exhibiting higher trust when information about certification was provided. However, expert users raised concerns about information overload, consistent with Kizilcec [38], who found that while information can enhance trust, overly detailed or complex disclosures may overwhelm users. Experts also questioned whether lay users could meaningfully process such details, aligning with research showing that certification information often exceeds users' comprehension capabilities [6, 14]. These insights highlight the need for layered transparency: high-level, accessible information for general users, complemented by more detailed documentation for experts. Additionally, integrating certification labels with explainability may enhance users' understanding, as suggested by recent research [17].

Both experts and non-experts preferred independent organizations, NGOs, or government agencies as certification bodies, which aligns with prior findings showing higher trust in non-profit or governmental certifiers than in for-profit entities [9]. Experts' observations that government authorities might lack AI expertise highlight a tension between non-profits and technical competence. As trust is defined as the willingness to be vulnerable based on expectations of competent performance [32], certification bodies must demonstrate both neutrality and expertise to earn user trust.

This suggests that AI certification should be overseen either by an independent or governmental authority with strong expertise, or through hybrid models that combine governmental neutrality with the knowledge of private companies.

Expert and non-expert users have fundamentally different mental models of the AI certification process. As expected, non-experts hold simpler models, while experts possess more sophisticated ones. These differences influence system design and communication. From an HCI perspective, this suggests that the certification process can retain technical complexity and multi-stakeholder interactions in the backend, but it should be presented to users in a simplified, easy-to-navigate, yet accurate manner to prevent misconceptions about how the AI certification process works.

Experts strongly emphasized context-specific certification standards, which highlight a fundamental challenge in AI governance: the tension between the desire for universal standards and the reality of AI systems' diverse applications and contexts. The expert insight that AI systems are "too diverse and complex for a one-size-fits-all approach" challenges simplified approaches to AI certification that attempt to apply uniform standards across vastly different use cases. This perspective is consistent with reports from industry practitioners who noted that appropriate evaluation metrics vary substantially across model architectures and use cases, and that existing regulations are often too abstract and underspecified to offer clear, domain-specific guidance for practical implementation [25].

Participants agreed certification should be ongoing. Non-experts favored regular, predictable reviews (e.g., annually), while experts preferred monitoring tied to system updates and retraining. This reflected different understandings of AI evolution: experts stressed rapid change, while non-experts thought in terms of slower-moving regulation. Similarly, industry professionals in a recent study [25] reported that AI governance policies are continually evolving, which complicates the timely review and implementation of required changes, and they emphasized a need for governance tools that support real-time monitoring and evaluation. Automated monitoring pipelines could help, though their adequacy remained uncertain.

Both user groups expressed concern about fraudulent AI certification and emphasized the need for protection against such fraud. However, the difference in their views on how to achieve this—with non-experts tending to focus on legal and regulatory measures and experts emphasizing technical solutions—suggests that addressing AI certification fraud is a complex challenge that requires multiple layers of safeguards. Another key consideration is how to make certification labels secure, verifiable, and resistant to counterfeiting. This is important because recent research has shown that users often struggle to distinguish authentic labels from misleading ones [16].

5.1 Recommendations for Improved AI Certification Schemes

We offer the following recommendations to enhance AI certification schemes and make them more user-centric. These recommendations are grounded in participant responses and address key HCI design challenges in making certification legible and actionable for diverse user populations. Because our study participants were EU

residents and frequently referenced EU regulations in their discussions, we use the EU AI Act as an illustrative comparison point; nonetheless, these recommendations contribute to the broader discourse on AI certification design. Future work should examine how our recommendations relate to the expectations of users outside the EU and to other regulatory frameworks worldwide, such as those in the US, UK, China, and Brazil.

- (1) **Enhance Transparency and Communication.** As preferred by both experts and non-experts, AI certificates should provide multi-level information. For non-experts, this means simple, easy-to-understand summaries that highlight main information like who issued the certification, how long it is valid, what aspects of the system were evaluated (e.g., safety, fairness, robustness), and simple system-level ratings (e.g., A, B, C) for each aspect. These ratings should be easy to interpret through a consistent visual design (e.g., color coding, icons) and interactive tooltips that explain each rating level. For experts, detailed technical documentation should be provided (e.g., using Model Cards [47]), including information on the training data, model architecture and design, evaluation metrics and methodologies used, performance results, and intended use cases. Communication should not only describe where the system performs well but also clearly explain where it may fail or involve risks (as noted by P_{E3} , P_{E11} , and P_{E14}). Any residual risks, such as untested edge cases, performance under rare conditions, or potential vulnerabilities, should be flagged with intuitive severity labels (e.g., low, medium, high) so users can quickly grasp their seriousness. In addition, contextual explanations using real-world examples, especially in high-risk domains like healthcare or autonomous driving, should be provided to help users understand how the system behaves in practice and form accurate mental models of what it can and cannot do. These user-centered design suggestions go beyond existing transparency requirements in the EU AI Act (Articles 11 and 16). They contribute to "nutrition-label-style" transparency and help calibrate user trust, guiding users to rely on the system appropriately without overestimating its capabilities. Therefore, these suggestions address trust calibration concerns raised by experts (e.g., as explained by P_{E7} , where "certified" may be misinterpreted as "risk-free").
- (2) **Require Independent and Multi-Stakeholder Certification Authorities.** Participants in both groups emphasized the importance of independent and credible certification authorities, citing concerns about bias and conflicts of interest. High-risk AI systems, in particular, should be certified by third-party accredited organizations rather than relying solely on self-assessment, although self-certification may be acceptable for lower-risk applications. To enhance legitimacy and robustness, hybrid certification models should be encouraged, involving NGOs, governmental agencies, and technical experts in collaborative processes. Additionally, user participation should be incorporated into certification schemes through mechanisms for feedback and issue reporting (as recommended by P_{E3} and P_{E7}). These recommendations reinforce the EU AI Act's provisions for conformity

assessment by notified bodies (Article 31) while extending accountability by advocating for stronger multi-stakeholder and user engagement.

- (3) **Adopt Flexible Certification.** Experts emphasized that certification must be context-specific and tailored to different use cases, noting that AI systems are too complex for binary yes-or-no labels. Certification should address specific aspects, as highlighted by participants such as P_{E5} and P_{E4} . This suggests that partial certification should be possible for particular features or contexts when full-system assessment is impractical. In line with Article 6 of the EU AI Act and the feedback from participants in both groups, certification approaches and standards should be adapted to the risk context, with high-risk AI systems requiring stricter standards, enhanced oversight, and more real-world testing.
- (4) **Implement Robust Post-Certification Monitoring.** Ongoing monitoring was seen as crucial by both experts and non-experts. As suggested by experts, certification schemes should require update-triggered monitoring, where significant system updates or retraining automatically prompt re-evaluation, especially for high-risk systems. Automated auditing pipelines could support compliance testing at scale, thus making monitoring more efficient in the context of rapidly evolving systems (as noted by P_{E11}). To enhance transparency and public trust, results of monitoring activities should be summarized and made accessible to users in accessible formats with visual aids rather than technical reports (as emphasized by non-experts like P_{N14}). These suggested practices align with the EU AI Act's requirements for continuous monitoring of high-risk AI systems (Article 21), but participants' expectations emphasized the importance of translating monitoring outcomes into information that is legible for lay users.
- (5) **Safeguard Certification Integrity and Prevent Fraud.** Participants expressed concerns about fraudulent certification. To maintain user trust in AI certification, several countermeasures addressing different aspects should be considered. First, as suggested by non-experts, legal enforcement mechanisms must be clearly defined, specifying penalties and remediation procedures for misuse, fraudulent claims, or non-compliance, while also allowing opportunities for corrective action before severe consequences are imposed. Second, as recommended by experts, technical safeguards and robust evaluation processes should be implemented to prevent fraud at any stage of the certification process. Third, cryptography-based approaches should be designed to make certification labels tamper-resistant, while ensuring that users can easily verify their authenticity with minimal effort and cognitive load. Furthermore, adopting one or a few widely recognized labels with clear, distinctive visual elements, supported by intuitive educational materials on identifying genuine certificates, can help reduce confusion and prevent misuse. Together, these measures directly address participants' fears of fraudulent practices, ensure reliable and trustworthy AI certification, and extend the compliance mechanisms outlined in the EU AI Act (Articles 16 and 31) by providing safeguards that are user-centric.

5.2 Limitations & Future Work

The limitations of our study are as follows: First, given its qualitative nature, we are unable to draw quantitative conclusions. Additionally, the study relies on self-reported data and assessments, which may be subject to biases such as social desirability, availability bias, and inaccurate recall or self-evaluation. Consequently, the data primarily reflect the highly subjective perspectives of the participants. Furthermore, cultural bias may have influenced the results. Despite our efforts to recruit a diverse sample, the study may still lack representativeness, limiting the generalizability of the findings. For instance, there is a generalization limitation for the expert group, as participants in this group are mainly from academia. Although many have AI industry experience from previous jobs or projects with industrial partners, this still limits the extent to which the findings reflect the views of experts working in companies. Moreover, all participants were either citizens or residents of WEIRD (Western, Educated, Industrialized, Rich and Democratic) countries, meaning their perceptions may be influenced by this. Individuals from non-WEIRD countries might hold different views regarding AI certification. Nonetheless, this exploratory study serves as an initial step in understanding the perceptions and expectations of both experts and non-experts regarding AI certification.

Future research should aim to include a more representative sample, including experts who are actively working in the industry as well as participants from a wider range of cultural and economic contexts. Additionally, it could explore perceptions of certifying human-AI systems in which humans and AI collaborate as integrated teams, for example in AI-assisted medical diagnosis or in collaborative robots used in manufacturing. Furthermore, future work could examine how users' perceived trustworthiness of certified AI systems compares to the actual trustworthiness of these systems.

6 Conclusion

In this study, we examined expert and non-expert end users' perceptions of AI certification to advance debates on AI regulation and inform the design of certification schemes that genuinely address user needs and support appropriate trust calibration. We conducted semi-structured interviews with 30 participants (15 experts and 15 non-experts) to explore their understanding of the certification process and their perspectives on post-certification monitoring, fraud risks, and prevention. Our user-centered investigation revealed significant differences between the two groups, as well as several critical challenges that demand attention from the HCI and AI governance communities. The varying expectations between experts and non-experts underscore the need for multi-level transparency and adaptive communication strategies in certification design. Moreover, static, one-size-fits-all certification approaches fail to address the dynamic, context-dependent nature of AI systems. We also identified challenges related to certification authority, fraud prevention, and ongoing monitoring that require coordinated efforts across technology development, and regulatory frameworks. Based on these findings, we offer actionable, user-centered recommendations for designing AI certification schemes that are both technically robust and socially relevant.

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